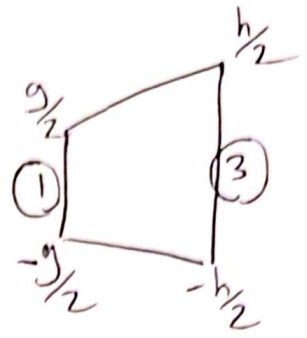


$$D = - \left[\iint \rho \mathbf{V} \cdot d\mathbf{s} u \right]$$

$$D = - \left[\int_{-h/2}^{h/2} \rho \mathbf{V} \cdot d\mathbf{s} u - \int_{-g/2}^{g/2} \rho \mathbf{V} \cdot d\mathbf{s} u \right]$$

$$D = - \left[\int_{-h/2}^{h/2} \rho u_3^2 dy - \int_{-g/2}^{g/2} \rho u_1^2 dy \right] \quad \text{Eq. (1) for the drag}$$



Continuity $\left[\int_{-g/2}^{g/2} \rho u_1 dy = \int_{-h/2}^{h/2} \rho u_3 dy \right] \times u_1$

$$\left[\int_{-g/2}^{g/2} \rho u_1^2 dy = \int_{-h/2}^{h/2} \rho u_3 u_1 dy \right] \quad (2)$$

replace $\int_{-g/2}^{g/2} \rho u_1^2 dy$ from (2) in (1), we get:

$$D = - \left[\int_{-h/2}^{h/2} \rho u_3^2 dy - \int_{-h/2}^{h/2} \rho u_3 u_1 dy \right] = - \left[\int_{-h/2}^{h/2} \rho u_3 (u_3 - u_1) dy \right] \quad \text{Eq. 3 for drag}$$

multiply continuity by u_3 , you get:

$$\int_{-g/2}^{g/2} \rho u_1 u_3 dy = \int_{-h/2}^{h/2} \rho u_3^2 dy \quad (4)$$

replace $\int_{-h/2}^{h/2} \rho u_3^2 dy$ from (4) in (1), we get:

$$D = - \left[\int_{-g/2}^{g/2} (\rho u_1 u_3 - \rho u_1^2) dy \right] = - \left[\int_{-g/2}^{g/2} \rho u_1 (u_3 - u_1) dy \right] \quad (5)$$

Eq. (1), (3) & (5) all give the same value for drag